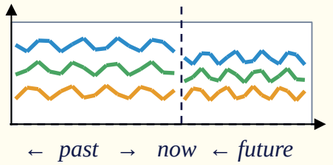
Time-Series Agents: From General to Specific

Scenario & Task Contract

Scenario

- Data streams & context
- Information boundary
- Operational objective



Define requirements

Task Contract

- Output & horizon
- Admissible information
- Constraints & metrics
- Review authority



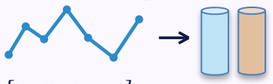
Select & specialize

Time-Series Agent

LLM-Side Capabilities §3

Temporal Representation & Alignment

Series ↔ language / multimodal representations



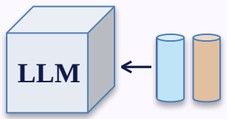
Temporal Understanding & Reasoning

Attributes · Instructions ·Reasoning



Model Adaptation

Training · Adapters · Supervision



LoRA Adapters Instruction supervision

Learned temporal capabilities

State-Actions Observations

General Harness (GH) §4

Reusable agent runtime

Interface & Interaction
State · Actions · Observations

Memory & Context
Store · Retrieve · Reuse

Tools & Execution
APIs · Code · Models · Simulators

Control, Verification & Feedback
Check · Retry · Recover

Planning & Workflow
Decompose · Schedule · Stop

Optimization & Evolution
Prompts · Tools ·Memory · Policies

Coordination & Scaling
Roles · Shared state ·Review

General Time-Series Harness (GTS) §5

Cross-dimensional temporal requirements

Temporal Causality
Event time · Availability time
As-of admissibility



Numerical Fidelity
Values · Scales · Units



Time-Indexed State Integrity
Alignment · Missingness · Provenance · Data vintage



Temporal Operation Semantics
Windowing · Frequency · Horizons



Temporal Grounding
Context · Lags · Evidence



Output Auditability
Traces · Checks · Review



Task-Specific Harness (TSK) §6

Task-bound outputs and validation

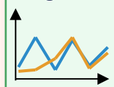
Forecasting & Prediction



Output: Future values / probabilities

Validation: Horizon · Backtesting · Calibration

Augmentation & Synthesis



Output: Generated time series

Validation: Temporal fidelity Downstream utility

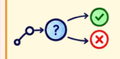
Anomaly Detection & Diagnosis



Output: Anomaly intervals / causes

Validation: Localization False alarms · Evidence

Decision Support



Output: Actions / recommendations

Validation: Utility · Risk Cost · Review

Evaluated by

Layer-Aligned Evaluation §7

Process Validity · GH
Tool-call validity · Recovery
Trace faithfulness
Memory / artifact provenance

Temporal Validity · GTS
As-of / leakage checks · Alignment
Temporal operations · Grounding
Units · Numerical fidelity

Task Performance · TSK
Forecast quality · Synthesis utility
Detection / diagnosis · Decision utility

Resource Behavior
Cost · Latency
Failures · Revisions

Auditable Episode Record
Task · As-of boundary · Visible context
Tool calls · Artifacts · Evidence
Validator decisions · Revisions
Final output · Resource use

Module-Level Evidence Ledger: Claims · Source evidence · Modules · Task contracts · Evaluation · Review status

Research Landscape of Time-Series Agent Systems

Model-side contributions, harness mechanisms, and evaluation resources

Model-Side Contributions

Representation · Alignment
Understanding · Adaptation

Representation & Alignment



- Time-LLM
- LLM4TS
- LLMTime
- ChatTS

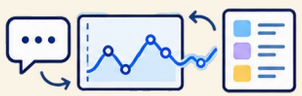
Learning & Reasoning



- Towards Time-Series Reasoning
- STRIDE
- OpenForecaster
- AURA-4B

Harness Mechanisms for Time-Series Agent Systems

1 Interface & Interaction



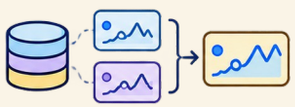
Structured Analysis Interfaces

- TS-Agent
- TimeART
- TSAG
- TESSA

Adaptive Interaction & Observation

- SAGE-Sleep
- ARTIST
- TimeRLM
- LLM-TSFD

2 Memory & Context



Retrieval & Context Construction

- Agentic RAG for TSA
- TimeCAP
- From News to Forecast
- ZARA
- MERIT

Temporal Experience & Evidence Memory

- FinMem
- MemCast
- ForecastCompass
- TIEM

3 Tools & Execution



Model & Code Execution

- TimeCopilot
- TimeSeriesScientist
- MoiraiAgent
- GenAutoML

Domain Tools & Simulation

- AgriWorld
- Open-TI
- GridMind
- PPG Physiological Agent

4 Control, Verification & Feedback



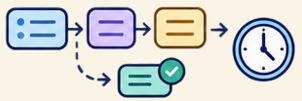
Review & Adjudication

- Argos
- AnomaMind
- TSQAgent
- Oil-Well Explainable Agent

Validation & Acceptance Gates

- Grid-Agent
- HydroAgent
- Neuro-Agentic Control
- BatteryLake

5 Planning & Workflow



Workflow Construction

- ClimateAgent
- DCATS
- MOSAIC
- Nexus

Operational Analysis Pipelines

- HFMD Structured Agents
- SIREN
- Smart Energy Agent
- S2S-FDD

6 Optimization & Evolution



Reflection & Experience Updates

- TimeClaw-EEL
- FLAIRR-TS
- CastFSR
- CRAFTER

Search & Adaptive Optimization

- MetaCaster
- EvoTS-Agent
- MarsTSC
- ATLAS

7 Coordination & Scaling



Debate & Evidence Aggregation

- TS-Debate
- MAS4TS
- Competition Agents
- FactorMAD

Coordinated Decision Systems

- FinCon
- TradingAgents
- CoLLMLight
- Fin-Analyst

Evaluation & Infrastructure

Benchmarks, datasets, simulators and task protocols for time-series agents

Benchmarks & Evaluation Resources



- TemporalBench
- TFRBench
- TraceBench
- TSAQA
- MMTS-Bench
- TimeSeriesGym
- TSQBench
- TimeSage-MT
- TS-Skill
- TimeSeriesExamAgent
- FutureSim
- WorldReasoner

Infrastructure & Task Protocols



- AION
- ModernTSF
- TimeClaw-Harness

Selected works grouped by relevant contributions. Subgroups provide descriptive reading aids; placements highlight selected mechanisms rather than exclusive paper categories.

Research routes toward recursive self-improvement

Selected milestones by first public year | Historical roots and the foundation-model era

Data & training	Tasks & curricula	Memory & skills	Code & workflows	Feedback & judges	Meta-level improvement
Synthetic data Persistent updates	Task generation Environment feedback	Reflective memory Reusable knowledge	Prompts and code Agent architectures	Learned rewards Evaluator learning	Update strategies Research automation

2026

- MetaClaw (Memory & skills)
- Prime Agent (Code & workflows)
- Meta^n (Meta-level improvement)
- Ind. Bilevel Autoresearch (Meta-level improvement)
- Hyperagents (Meta-level improvement)

2025

- SEAL (Data & training)
- AgentEvolver (Tasks & curricula)
- ACE (Memory & skills)
- GEPA (Code & workflows)
- AlphaEvolve (Meta-level improvement)
- R-Zero (Tasks & curricula)
- Dynamic Cheatsheet (Memory & skills)
- DGM (Code & workflows)
- Process Self-Rewarding (Feedback & judges)
- TTRL (Data & training)
- Absolute Zero (Tasks & curricula)
- A-MEM (Memory & skills)
- SICA (Code & workflows)
- AI Scientist-v2 (Meta-level improvement)

2024

- SCoRe (Data & training)
- RISE (Data & training)
- Quiet-STaR (Data & training)
- SPIN (Data & training)
- AgentGym (Tasks & curricula)
- AFlow (Code & workflows)
- CREAM (Feedback & judges)
- Self-Taught Evaluators (Feedback & judges)
- Meta-Rewarding (Feedback & judges)
- Self-Rewarding (Feedback & judges)
- ADAS (Code & workflows)
- TextGrad (Code & workflows)
- AI Scientist (Meta-level improvement)

2023

- ReST^{EM} (Data & training)
- ReST (Data & training)
- ExpeL (Memory & skills)
- Voyager (Memory & skills)
- Reflexion (Memory & skills)
- DSPy (Code & workflows)
- OPRO (Code & workflows)
- CRITIC (Feedback & judges)
- STOP (Meta-level improvement)
- Promptbreeder (Meta-level improvement)

2022

- Self-Instruct (Data & training)
- STaR (Data & training)
- Constitutional AI (Feedback & judges)

Historical roots

- **Gödel Machine** — 2003 | Provable self-modification
- **Learned optimizers** — 2016 | Learning update rules
- **POET** — 2019 | Tasks and solvers co-evolve

Marks: selected affiliations | Ind.: independent researchers | Branches: research routes, not technical ancestry